Laboratory work 5

Task1: Add a CHECK constraint to passenger table to provide that passengers must be at least 10 years old.

```
ALTER TABLE passengers
ADD CONSTRAINT check_age_10
CHECK (
    date_of_birth <= CURRENT_DATE - INTERVAL
'10 years'
);
```

```
15
16  --Task 1. Passengers must be at least 10 years old
17  ALTER TABLE passengers
18  ADD CONSTRAINT check_age_10
19  CHECK (
20      date_of_birth <= CURRENT_DATE - INTERVAL '10 years'
21  );
22
```

Data Output | Messages | Notifications

```
ALTER TABLE

Query returned successfully in 107 msec.
```

- -- Task 2
- -- Booking price must be between 0 and 50000

- ALTER TABLE booking
- ADD CONSTRAINT check_booking_price
- CHECK (price BETWEEN 0 AND 50000);

```
22  -- Task 2.Booking price must be between 0 and 50000
23
24  ALTER TABLE booking
25  ADD CONSTRAINT check_booking_price
26  CHECK (price BETWEEN 0 AND 50000);
```

Data Output | Messages | Notifications

```
ALTER TABLE

Query returned successfully in 2 secs 474 msec.
```

Task3.Add a CHECK constraint to accept the luggage weight between 1 and 23 kg.

ALTER TABLE baggage

ADD CONSTRAINT check_baggage_weight

CHECK (weight_in_kg BETWEEN 1 AND 23);

```
34
35  -- Task 3. Luggage weight between 1 and 23 kg
36
37  ALTER TABLE baggage
38  ADD CONSTRAINT check_baggage_weight
39  CHECK (weight_in_kg BETWEEN 1 AND 23)
40  NOT VALID;
41
42
43
44  -- Check Task 3 constraint
45
46  SELECT
47      conname,
48      pg_get_constraintdef(oid)
49  FROM pg_constraint
50  WHERE conrelid = 'baggage'::regclass
51  AND conname = 'check_baggage_weight';
```

Data Output | Messages | Notifications

SQL | Showing rows: 1 to 1

	conname name	pg_get_constraintdef text
1	check_baggage_wei...	CHECK (((weight_in_kg >= (1)::numeric) AND (weight_in_kg <= (23)::numeric))) NOT V...

Task 4.Airport name must have at least 10 characters

ALTER TABLE airport

ADD CONSTRAINT check_airport_name

CHECK (LENGTH(airport_name) >= 10);

```
53
54  -- Task 4.Airport name must have at least 10 characters
55
56  ALTER TABLE airport
57  ADD CONSTRAINT check_airport_name
58  CHECK (LENGTH(airport_name) >= 10);
59
```

Data Output | Messages | Notifications

```
ALTER TABLE

Query returned successfully in 2 secs 575 msec.
```

-- Task 5.Add UNIQUE constraints

```
-- 1. Passengers
ALTER TABLE passengers
ADD CONSTRAINT unique_passport_number
UNIQUE (passport_number);
-- 2. Airline
ALTER TABLE airline
ADD CONSTRAINT unique_airline_code
UNIQUE (airline_code);
        -- 3. Airport
ALTER TABLE airport
ADD CONSTRAINT unique_airport_location
UNIQUE (airport_name, city, country);
      -- 4. Baggage
ALTER TABLE baggage
ADD CONSTRAINT unique_baggage_booking
UNIQUE (baggage_id, booking_id);
  -- 5. Baggage Check
ALTER TABLE baggage_check
ADD CONSTRAINT unique_baggage_check_record
UNIQUE (booking_id, passenger_id, created_at);
-- 6. Boarding Pass
ALTER TABLE boarding_pass
ADD CONSTRAINT unique_boarding_pass_booking_seat
UNIQUE (booking_id, seat);
```

Data Output | Messages | Notifications

Showing rows

	table_name regclass	constraint_name name	definition text
1	airline	unique_airline_code	UNIQUE (airline_code)
2	airport	unique_airport_location	UNIQUE (airport_name, city, country)
3	baggage_ch...	unique_baggage_check_record	UNIQUE (booking_id, passenger_id, created_at)
4	baggage	unique_baggage_booking	UNIQUE (baggage_id, booking_id)
5	boarding_pa...	unique_boarding_pass_booking_s...	UNIQUE (booking_id, seat)
6	passengers	unique_passport_number	UNIQUE (passport_number)
7	security_che...	unique_security_check_record	UNIQUE (passenger_id, created_at, check_res...

```
-- 7. Security Check
-- Check for duplicates before adding
ALTER TABLE security_check
ADD CONSTRAINT unique_security_check_record
UNIQUE (passenger_id, created_at, check_result);
```

Task 6 — CHECK Constraint Based on Gender

```
277
278  -- Task 6. Age restriction based on gender
279
280  ALTER TABLE passengers
281  ADD CONSTRAINT check_gender_age
282  CHECK (
283      (gender = 'Male'
284       AND date_of_birth <= CURRENT_DATE - INTERVAL '18 years')
285      OR
286      (gender = 'Female'
287       AND date_of_birth <= CURRENT_DATE - INTERVAL '19 years')
288  )
289  NOT VALID;
290
291  -- Task 6. Check constraint result
292
293  SELECT
294      conname,
295      convalidated,
296      pg_get_constraintdef(oid)
297  FROM pg_constraint
298  WHERE conrelid = 'passengers'::regclass
299  AND conname = 'check_gender_age';
300
```

Data Output | Messages | Notifications

Showing rows: 1 to 1 | Page N

	conname name	convalidated boolean	pg_get_constraintdef text
1	check_gender_a...	false	CHECK (((((gender)::text = 'Male'::text) AND (date_of_birth <= (CURRENT_DATE - '18 years'::interval))) OR

Task 7 — Age restriction based on citizenship

Query | Query History

```
305 ALTER TABLE passengers
306 ADD CONSTRAINT check_country_age
307 CHECK (
308     (country_of_citizenship = 'Kazakhstan'
309      AND date_of_birth <= CURRENT_DATE - INTERVAL '18 years')
310     OR
311     (country_of_citizenship = 'France'
312      AND date_of_birth <= CURRENT_DATE - INTERVAL '17 years')
313     OR
314     (country_of_citizenship NOT IN ('Kazakhstan', 'France')
315      AND date_of_birth <= CURRENT_DATE - INTERVAL '19 years')
316 )
317 NOT VALID;
318
319
320 -- Task 7. Check constraint result
321
322 SELECT
323     conname,
324     convalidated,
325     pg_get_constraintdef(oid)
326 FROM pg_constraint
327 WHERE conrelid = 'passengers'::regclass
328 AND conname = 'check_country_age';
329
```

Data Output | Messages | Notifications

Showing rows: 1 to 1 | Page No

	conname name	convalidated boolean	pg_get_constraintdef text
1	check_country_a...	false	CHECK (((((country_of_citizenship)::text = 'Kazakhstan'::text) AND (date_of_birth <= (CURRENT_DATE - '18

Task 8 — Ticket Discount

```
-- Task 8. Add ticket_discount column.
ALTER TABLE booking
ADD COLUMN ticket_discount NUMERIC;
-- Task 8. Update ticket discounts
UPDATE booking
SET ticket_discount =
    CASE
        WHEN created_at > '2024-01-01' THEN 5
        WHEN created_at < '2024-01-01' THEN 10
        ELSE 0
    END;
-- Task 8. Add CHECK constraint
ALTER TABLE booking
ADD CONSTRAINT check_ticket_discount
CHECK (
    (created_at > '2024-01-01' AND ticket_discount = 5)
    OR
    (created_at < '2024-01-01' AND ticket_discount = 10)
    OR
    (created_at = '2024-01-01' AND ticket_discount = 0)
);
```

```
-- Task 8. Check results

SELECT
    booking_id,
    price,
    created_at,
    ticket_discount
FROM booking
LIMIT 10;
```

```
347
348  -- Task 8. Add CHECK constraint
349
350  ALTER TABLE booking
351  ADD CONSTRAINT check_ticket_discount
352  CHECK (
353      (created_at > '2024-01-01' AND ticket_discount = 5)
354      OR
355      (created_at < '2024-01-01' AND ticket_discount = 10)
356      OR
357      (created_at = '2024-01-01' AND ticket_discount = 0)
358  );
359
```

Data Output | Messages | Notifications

SQL

	booking_id [PK] integer	price numeric (7,2)	created_at date	ticket_discount numeric
1	1	7462.13	2023-06-06	10
2	2	4216.60	2024-01-03	5
3	3	5782.37	2024-01-12	5
4	4	102.96	2024-01-31	5
5	5	6711.04	2023-04-25	10
6	6	7813.55	2023-12-23	10
7	7	1346.71	2024-02-28	5
8	8	5711.67	2023-10-06	10
9	9	7015.82	2023-12-28	10
10	10	1668.38	2024-01-07	5

Task 9 — DEFAULT Booking Status

```sql

-- Task 9. Set default booking status

ALTER TABLE booking
ALTER COLUMN status
SET DEFAULT 'Confirmed';


-- Task 9. Check DEFAULT value

SELECT
    column_name,
    column_default
FROM information_schema.columns
WHERE table_name = 'booking'
AND column_name = 'status';

```

Data Output | Messages | Notifications

	column_name name	column_default character varying
1	status	'Confirmed'::character varyi...

Query | Query History

```sql
-- Task 9. Insert booking without status

INSERT INTO booking (
    booking_id,
    passenger_id,
    booking_platform,
    created_at,
    price,
    ticket_discount
)
VALUES (
    (SELECT MAX(booking_id) + 1 FROM booking),
    1,
    'Website',
    CURRENT_DATE,
    15000,
    5
);


-- Task 9. Check inserted booking
```

Data Output | Messages | Notifications

Showing

	booking_id [PK] integer	passenger_id integer	status character varying (50)	price numeric (7,2)	ticket_discount numeric
1	501	1	Confirmed	15000.00	5
2	500	130	Female	950.80	10
3	499	169	Male	1411.72	10
4	498	35	Female	9424.91	10
5	497	118	Male	734.80	10

Task 10 — Automatic Current Date

```sql

-- Task 10. Set automatic current date

ALTER TABLE booking
ALTER COLUMN created_at
SET DEFAULT CURRENT_DATE;


-- Task 10. Check DEFAULT value

SELECT
    column_name,
    column_default
FROM information_schema.columns
WHERE table_name = 'booking'
AND column_name = 'created_at';
```

Data Output | Messages | Notifications

	column_name name	column_default character varying
1	created_at	CURRENT_DATE

Query | Query History

```sql
-- Task 10. Insert booking without created_at

INSERT INTO booking (
    booking_id,
    passenger_id,
    booking_platform,
    price,
    ticket_discount
)
VALUES (
    (SELECT MAX(booking_id) + 1 FROM booking),
    1,
    'Website',
    12000,
    5
);


-- Task 10. Check automatic date

SELECT
    booking_id,
    passenger_id,
    created_at,
```

Data Output | Messages | Notifications

	booking_id [PK] integer	passenger_id integer	created_at date	status character varying (50)	price numeric (7,2)
1	502	1	2026-10-10	Confirmed	12000.00
2	501	1	2026-10-10	Confirmed	15000.00
3	500	130	2023-10-13	Female	950.80